5.2 Similarity Theorems (AA, SAS~, SSS~)

Similar triangles have the same shape but not necessarily the same size: corresponding angles are congruent, and corresponding sides are in a constant ratio (the scale factor). Three shortcuts prove similarity without checking every part. **AA (Angle-Angle):** two pairs of congruent angles are enough, since the third pair automatically matches (Angle-Sum Theorem). **SAS~ (Side-Angle-Side):** two pairs of sides in the same ratio, with the included angle congruent. **SSS~ (Side-Side-Side):** all three pairs of corresponding sides in the same ratio. A key skill is matching corresponding sides correctly — often this means sorting each triangle's sides from shortest to longest before comparing ratios. Once similarity is established, a proportion between corresponding sides (written in the order given by the similarity statement) solves for missing lengths, and perimeters of similar triangles share the same ratio as their sides.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Two triangles each have angles 55° and 70°. Are they similar? Why?

2. Sides 4, 6, 8 and 6, 9, 12. Similar?

3. $\triangle ABC \sim \triangle DEF, AB = 5, DE = 15, BC = 7$. Find EF.

4. Corresponding sides 8 and 2. Scale factor from larger to smaller?

5. Sides 3, 5 with included 60°; sides 6, 10 with included 60°. Similar?

6. $\triangle GHI \sim \triangle JKL, GH = 10, JK = 4, HI = 15$. Find KL.

7. Sides 5, 7, 9 and 10, 14, 20. Similar?

8. $\overline{DE} \parallel \overline{BC}$ in $\triangle ABC, AD = 4, AB = 10, DE = 6$. Find BC.

9. $\angle A \cong \angle D$ and $\angle B \cong \angle E$. Write a similarity statement.

10. $\triangle RST \sim \triangle UVW, RS = 9, UV = 12, ST = 15$. Find VW.

11. Sides 6, 8, 10 correspond to 9, 12, 15. Scale factor?

12. Angles 30°, 80° in one triangle; 30°, 70° in another. Similar?

NAME: _____ Geometry (FL B.E.S.T.)

13. In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$, $AD = 3x$, $DB = x + 2$, $AE = 9$, $EC = 6$, and $\triangle ADE \sim \triangle ABC$. Find x.

14. $\triangle PQR \sim \triangle XYZ$. $PQ = 7$, $QR = 9$, $RP = 11$, and $XY = 21$. Find the scale factor, YZ, ZX, and the perimeter of $\triangle XYZ$.

15. A triangle with sides x, 12, 15 (where x is shortest) is similar to a triangle with sides 8, 20, 25. Find x.

✓ Answer Key

1. Yes; AA (matching angle pairs 55° and 70°)
2. Yes; all ratios equal $\frac{3}{2}$
3. 21
4. $\frac{1}{4}$
5. Yes; SAS~
6. 6
7. No; $\frac{20}{9} \neq 2$ (the other two ratios equal 2)
8. 15
9. $\triangle ABC \sim \triangle DEF$
10. 20
11. $\frac{3}{2}$
12. No; third angles differ (70° vs 80°)
13. $x = 2$
14. Scale factor = 3; $YZ = 27$, $ZX = 33$; perimeter of $\triangle XYZ = 81$
15. $x = 4.8$